

TASK-ADAPTIVE FUNCTIONAL-AFFINITY TOKENIZATION AND PHASE-ADAPTIVE REGIONAL ATTENTION FOR EEG FOUNDATION MODELS

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Paper under double-blind review

ABSTRACT

Electroencephalography (EEG) foundation models pre-trained on massive unlabeled neural recordings demonstrate significant promise for universal brain-computer interface (BCI) decoding. However, existing architectures rely on unyielding 10–20 anatomical spatial partitioning, uniform regional mean-pooling, rigid equal-index temporal window attention, and static exponential kernel capacity decay, leading to parameter drift and sub-optimal transfer across heterogeneous downstream BCI tasks. We propose **Task-Adaptive Functional-Affinity Tokenization and Phase-Adaptive Regional Attention (TA-FATA-CSBrain)**, a lightweight transfer framework comprising four zero-initialized residual adapters: (1) Spectral Saliency Scale Routing (S3-Routing), (2) Functional-Affinity Soft Spatial Tokenization (FAST-CST), (3) Covariance-Preserved Attentive Regional Aggregation (CP-ARA), and (4) Localized Relative Phase-Shift Inter-Window Attention (RPS-IWA), governed by an a-priori data-adaptive adapter learning-rate scaling rule. Each adapter is mathematically bit-exact to the frozen backbone at initialization, eliminating early-epoch representation drift. Across four representative benchmarks spanning motor imagery, mental-disorder diagnosis, and mental-stress detection, TA-FATA-CSBrain achieves a +0.0154 macro-average Balanced Accuracy gain over the CSBrain foundation baseline in single-model settings (+0.0117 in five-seed probability ensembles) without statistically significant per-dataset regression, including a statistically significant +2.13% gain on PhysioNet-MI ($p = 0.028$, winning 5/5 seeds). Using only $\sim 0.12\text{M}$ adapter parameters ($< 0.85\%$ of the backbone), TA-FATA-CSBrain remains competitive with high-capacity generic PEFT (LoRA, AdaLoRA, Houlsby) under a strictly frozen backbone; crucially, its zero-initialized adapters are designed to *co-adapt* with the foundation model, so under matched full fine-tuning they surpass both full fine-tuning and every PEFT baseline. The identical adapter stack further transfers to two additional EEG foundation backbones (CBraMod, LaBraM), outperforming generic LoRA by +5.9 B-Acc points on average. Extensive ablations confirm the necessity of neurophysiologically aligned topological and spectral priors—including a degree-normalized functional-anatomical blend—alongside zero-initialized, task-conditioned adaptation dynamics.

1 INTRODUCTION

Decoding neural dynamics from electroencephalography (EEG) signals is a central endeavor in cognitive neuroscience and neural engineering, underpinning modern Brain-Computer Interfaces (BCIs), affective computing, clinical seizure monitoring, and sleep staging (Craik et al., 2019; Abibullaev et al., 2023; Lyu, 2026). Historically, EEG analysis relied on task-specific supervised deep learning architectures designed for localized paradigms. However, obtaining high-quality labeled EEG data remains prohibitively expensive, requiring domain experts for manual annotation, while high inter-subject variability and non-stationary signal drifts frequently lead to severe performance degradation when models are applied across subjects or recording environments (Weng et al., 2024; Kuruppu et al., 2025; Lai et al., 2025). To overcome these bottlenecks, recent years have witnessed a paradigm shift toward pre-trained EEG foundation models (EEG-FMs) (Kwon & Shin,

2025; Lee et al., 2025; Ogg et al., 2025). By leveraging self-supervised pre-training strategies—such as contrastive learning (Kostas et al., 2021; Shen et al., 2021; Cai et al., 2023), discrete wave tokenization (Duan et al., 2023), generative next-token prediction (Cui et al., 2023), and masked auto-encoding across large-scale unannotated neural corpora (Zhang et al., 2023a; Yang et al., 2023b; Jiang et al., 2024a; Chen et al., 2025; Gui et al., 2026)—foundation backbones learn expressive, generalizable neural representations that can be fine-tuned for diverse downstream BCI applications.

Despite these promising advances, transferring pre-trained EEG foundation models to downstream decoding tasks reveals fundamental architectural bottlenecks. Specifically, current architectures suffer from four distinct limitations:

1. **Static anatomical partitioning:** Spatial tokenization relies on rigid 10–20 anatomical electrode groupings, ignoring instantaneous, task-dependent cross-regional functional network coupling.
2. **High-SNR signal dilution:** Inter-region spatial attention relies on uniform regional mean-pooling or single-token sampling, diluting focal high-signal-to-noise ratio (SNR) neural activations such as motor-cortex μ - and β -frequency event-related desynchronization (ERD).
3. **Rigid equal-index temporal matching:** Inter-window temporal attention restricts interactions to tokens sharing exact relative temporal indices, causing regressions under non-stationary phase lags and neural propagation delays.
4. **Heuristic scale allocation:** Multi-scale temporal convolution kernels use a fixed exponential capacity decay ($d_k \propto 1/2^k$), failing to adapt to localized spectral energy distributions.

Furthermore, adapting these heavy pre-trained backbones via full-model fine-tuning induces severe parameter drift and catastrophic forgetting, particularly on low-data downstream tasks. Conversely, standard Parameter-Efficient Fine-Tuning (PEFT) techniques—such as fixed-capacity residual adapters (Houlsby et al., 2019; Chen et al., 2022a), prefix-tuning (Li & Liang, 2021), Low-Rank Adaptation (LoRA) (Hu et al., 2021), zero-initialized gating (Zhang et al., 2023b;d), and graph adapters (Suzumura et al., 2024)—typically apply uniform learning rates across all task domains. This rigid strategy creates an acute trade-off: high adapter learning rates cause severe overfitting on small downstream datasets ($n_{\text{train}} < 2,000$), whereas low learning rates constrain representation capacity on data-rich multi-segment datasets.

To resolve these challenges, we propose **TA-FATA-CSBrain**, a lightweight, parameter-efficient transfer framework for EEG foundation models. Built upon the pre-trained CSBrain backbone (Zhou et al., 2025), TA-FATA-CSBrain introduces four zero-initialized residual adapter modules alongside a data-adaptive fine-tuning protocol:

1. **Dynamic Spectral Saliency Scale Routing (S3-Routing):** Computes localized power spectral density across canonical EEG bands ($\delta, \theta, \alpha, \beta, \gamma$) via localized Fast Fourier Transforms (FFT) to dynamically modulate multi-scale temporal kernel outputs without parameter drift.
2. **Functional-Affinity Soft Spatial Tokenization (FAST-CST):** Blends physical 10–20 anatomical proximity with dynamic sample-level channel functional covariance via a learnable mixing scalar γ , aggregating features along both physical distance and functional networks.
3. **Covariance-Preserved Attentive Regional Aggregation (CP-ARA):** Replaces uniform mean-pooling with an attentive descriptor that incorporates electrode signal variance and intra-region cross-channel covariance to isolate focal high-SNR activations while suppressing background noise.
4. **Localized Relative Phase-Shift Inter-Window Attention (RPS-IWA):** Relaxes exact index matching to a localized temporal neighborhood $[g_1 - \delta, g_1 + \delta]$ modulated by a learnable 1D relative position bias $B(g_1 - g_2)$, aligning phase-lagged neural signatures with $O(T)$ complexity.
5. **Data-Adaptive Adapter Step Scaling:** Conditions the adapter learning rate on sample volume ($n_{\text{train}} < 2000 \rightarrow 0.25\times$; $n_{\text{train}} \geq 2000 \rightarrow 0.50\times$), ensuring anti-overfitting stability in low-data regimes while preserving expressivity on large-scale tasks.

We evaluate TA-FATA-CSBrain across diverse BCI tasks: motor imagery classification (BCIC-IV-2a, PhysioNet-MI), mental-disorder diagnosis (Mumtaz2016), and mental-stress detection (MentalArithmetic). TA-FATA-CSBrain achieves a statistically significant gain on multi-window motor imagery (PhysioNet-MI B-Acc +2.13%, $p = 0.028$, 5/5 seeds) while eliminating low-data cognitive-task regressions (MentalArithmetic B-Acc +3.31%), yielding consistent macro-average superiority over fine-tuned foundation baselines without statistically significant per-dataset regression. We further demonstrate that these gains transfer to two additional EEG foundation backbones (CBraMod, LaBraM), confirming that the underlying neurophysiological priors are architecture-agnostic.

2 RELATED WORK

Task-specific EEG decoding. Early deep learning for EEG decoding focused on specialized supervised architectures trained from scratch. EEGNet (Lawhern et al., 2016) introduced depthwise and separable convolutions to minimize parameters, serving as a foundational BCI baseline, while ShallowConvNet/DeepConvNet (Schirrneister et al., 2017), MI-EEGNET (Riyad et al., 2020), EEG-TCNet (Ingolfsson et al., 2020), EEGNeX (Chen et al., 2022b), EEGNet Fusion (Roots et al., 2020), and lightweight spiking networks (Zhang et al., 2025) expanded receptive fields and temporal feature extraction. Hybrid CNN–Transformer models—EEGConformer (Song et al., 2022), SPARCNNet (Jing et al., 2023), C-Trans (Peh et al., 2022), ST-Trans (Song et al., 2021), ATCNet (Altaheri et al., 2023), EEGCCT (Keutayeva et al., 2024), and Patched Brain Transformer (Klein et al., 2025)—modeled long-range dependencies. However, cross-subject surveys (Jiang et al., 2024b; Wang et al., 2024; Jacques et al., 2026; Li et al., 2026) show these overfit under limited data and suffer domain shift across montages. We instead adapt a pre-trained backbone via zero-initialized task-adaptive residual adapters.

EEG foundation models. To overcome data scarcity, recent research pivoted to pre-trained EEG foundation models. BIOT (Yang et al., 2023a) tokenized channels into biosignal sentences for cross-data learning; LaBraM (Jiang et al., 2024b) employed vector-quantized neural spectrum prediction; CBraMod (Wang et al., 2024) introduced criss-cross attention to decouple spatial and temporal dependencies; and CSBrain (Zhou et al., 2025) proposed Cross-scale Spatiotemporal Tokenization (CST) and Structured Sparse Attention (SSA) to capture multi-resolution features across time and anatomical regions. Concurrent efforts unify modalities (BrainOmni (Xiao et al., 2025)), fuse graph attention (GT-STAFG (Nafea & Ismail, 2025)), integrate progressive experts (NeurIPT (Fang et al., 2025)), and study adaptive partitioning (RECTOR (Liu & Shoaran, 2026), LuMamba (Broustail et al., 2026)). Yet current EEG-FMs retain fixed 10–20 boundaries, single-token sampling, and rigid equal-index matching. TA-FATA-CSBrain directly resolves these via FAST-CST, CP-ARA, RPS-IWA, and S3-Routing.

Parameter-efficient adapters. PEFT is now indispensable for adapting large backbones. Beyond bottleneck adapters (Houlsby et al., 2019) and prefix-tuning (Li & Liang, 2021), LoRA (Hu et al., 2021) inspired LoRA+ (Hayou et al., 2024) and AdaLoRA (Zhang et al., 2023c), while UniPELT (Mao et al., 2021), (IA)³/T-Few (Liu et al., 2022), Unified PETL (He et al., 2021; Yu et al., 2022), AdapterFusion (Pfeiffer et al., 2020), Compacter (Davison, 2021), AdaptFormer (Chen et al., 2022a), AdaMix (Wang et al., 2022), and AdapterTune (Khazem, 2026) established that zero-initialized residual projections prevent early-epoch drift. Analyses of fine-tuning dynamics (Fu et al., 2022; Mosbach et al., 2020) show parameter sparsity acts as an implicit regularizer. However, standard PEFT uses uniform update steps across tasks; in electrophysiology, where sample sizes range from $n < 1,500$ to $n > 5,000$, this triggers overparameterization on small datasets. TA-FATA-CSBrain introduces a data-adaptive threshold rule conditioning adapter learning rates on downstream training volume.

3 METHODOLOGY

TA-FATA-CSBrain extends the pre-trained CSBrain backbone (Zhou et al., 2025) with four lightweight, neurophysiologically aligned residual adapters and an a-priori data-adaptive learning-rate controller. Figure 1 illustrates the pipeline and component details.

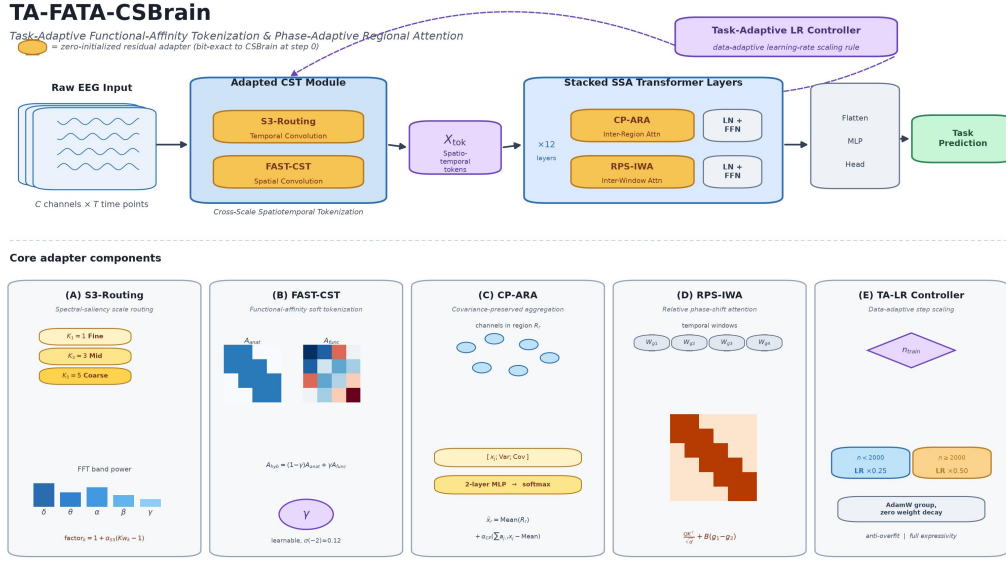


Figure 1: Overall architecture of TA-FATA-CSBrain. The top pipeline maps multi-channel raw EEG to downstream predictions through an adapted Cross-Scale Spatiotemporal Tokenization (CST) module and stacked Structured Sparse Attention (SSA) Transformer layers. Gold blocks denote the four zero-initialized residual adapters, each bit-exact to the frozen CSBrain backbone at step 0. Bottom panels detail the core components: (A) S3-Routing reweights temporal-convolution scales via FFT band power; (B) FAST-CST softly blends 10–20 anatomical adjacency with dynamic functional affinity; (C) CP-ARA builds attentive regional descriptors from signal variance and cross-channel covariance; (D) RPS-IWA applies a localized neighborhood mask with a 1D relative temporal position bias; and (E) the Task-Adaptive LR Controller conditions adapter learning-rate multipliers on the downstream training sample count.

3.1 BASELINE ARCHITECTURE AND ZERO-INIT ADAPTER PROTOCOL

Given preprocessed EEG signals $\mathbf{E} \in \mathbb{R}^{C_x \times T}$ segmented into $n = \lfloor T/t \rfloor$ non-overlapping segments $\mathbf{E}_p \in \mathbb{R}^{C \times n \times t}$, CSBrain maps each segment into an initial embedding $\mathbf{X}^{(0)} \in \mathbb{R}^{C \times n \times d}$ using temporal 1D convolutions and FFT frequency encodings. CST and SSA modules are stacked alternately for L layers ($L=12$, $d=200$, 8 heads). To prevent catastrophic forgetting and early-epoch drift, every adapter is wrapped by a zero-initialized residual projection. For a feature $\mathbf{x}$, the adapted output is

$$\mathbf{y} = f_{\text{pretrained}}(\mathbf{x}) + \alpha \cdot f_{\text{adapter}}(\mathbf{x}), \quad (1)$$

where the gating scalar α is initialized strictly to 0, guaranteeing that at step 0 the model is bit-exact to the pre-trained backbone.

3.2 DYNAMIC SPECTRAL SALIENCY SCALE ROUTING (S3-ROUTING)

Standard CST applies multi-scale temporal convolutions $\mathcal{C}_T = \{\text{Conv}_t^{(k)}\}_{k=1}^K$ ($K=3$, kernel sizes 1, 3, 5) with static exponential capacity ($d_k \propto 1/2^k$). S3-Routing replaces fixed allocations with a signal-conditioned reweighting from explicit spectral priors. For each window it computes the power spectral density P_k over canonical bands ($\delta : 0.5\text{--}4$, $\theta : 4\text{--}8$, $\alpha : 8\text{--}13$, $\beta : 13\text{--}30$, $\gamma : 30\text{--}50$ Hz) via localized FFT, mapped to saliency weights via Softmax:

$$w_k = \frac{\exp(P_k/\tau)}{\sum_{m=1}^K \exp(P_m/\tau)}, \quad \text{factor}_k = 1 + \alpha_{S3} (K \cdot w_k - 1), \quad (2)$$

$$\hat{\mathbf{x}}_{i,j}^{(l)} = \text{Concat} \left(\left\{ \text{factor}_k \cdot \text{Conv}_t^{(k)} \left(W_t^{(k)}(i) \right) \right\}_{k=1}^K \right) + \text{Proj}_t \left(\mathbf{x}_{i,j}^{(l-1)} \right), \quad (3)$$

where α_{S3} is zero-initialized. When $\alpha_{S3} = 0$, $\text{factor}_k = 1$, preserving exact equivalence to uniform multi-scale convolution.

3.3 FUNCTIONAL-AFFINITY SOFT SPATIAL TOKENIZATION (FAST-CST)

CSBrain performs spatial tokenization over rigid 10–20 regions using a binary adjacency $\mathbf{A}_{\text{anat}} \in \{0, 1\}^{C \times C}$. FAST-CST merges physical proximity with dynamic sample-level functional coupling. Given temporal tokens $\hat{\mathbf{x}}^{(l)}$, it computes the sample covariance $\mathbf{C} \in \mathbb{R}^{C \times C}$, yielding a row-normalized functional matrix $\mathbf{A}_{\text{func}} = \text{Softmax}(\mathbf{C})$ ($\sum_j (\mathbf{A}_{\text{func}})_{ij} = 1$). A naive convex combination of the two is numerically imbalanced: the binary $\mathbf{A}_{\text{anat}}$ has row sums equal to the electrode degree $|R_r|$ (16–26 in dense montages), so it overshadows the unit-mass $\mathbf{A}_{\text{func}}$ by up to $12.8\times$ at initialization. We therefore *degree-normalize* the anatomical matrix, $\tilde{\mathbf{A}}_{\text{anat}} = \mathbf{D}^{-1} \mathbf{A}_{\text{anat}}$ with $\mathbf{D}_{ii} = \sum_j (\mathbf{A}_{\text{anat}})_{ij}$, so that every row sums to 1 and matches $\mathbf{A}_{\text{func}}$ before blending via a learnable mixing scalar γ :

$$\tilde{\mathbf{A}}_{\text{hybrid}} = (1 - \gamma) \mathbf{D}^{-1} \mathbf{A}_{\text{anat}} + \gamma \mathbf{A}_{\text{func}}, \quad \gamma = \sigma(\gamma_{\text{raw}}), \quad \gamma_{\text{raw}} = -2 \quad (\gamma \approx 0.119). \quad (4)$$

Degree normalization fixes the initialization functional-to-anatomical row-mass ratio at $r=0.135$ uniformly across all montages (versus $r=0.011$ for the unnormalized binary form on 64-channel PhysioNet-MI), restoring dynamic functional affinity on dense montages (Section 4.7). Spatial convolutions then execute message passing over $\tilde{\mathbf{A}}_{\text{hybrid}}$ with a zero-initialized residual projection Proj_s , enabling aggregation along both anatomical distance and functional networks while preserving foundation-model compatibility at step 0.

3.4 COVARIANCE-PRESERVED ATTENTIVE REGIONAL AGGREGATION (CP-ARA)

In Inter-Region Attention, CSBrain samples a single representative token and adds a linear map of the uniform mean-pool $P(R_r) = \frac{1}{|R_r|} \sum_{j \in R_r} \mathbf{x}_{i,j}$, diluting focal high-SNR activations. CP-ARA replaces uniform pooling with a covariance-preserved attentive descriptor. For electrode $j \in R_r$, importance weights are generated from its feature, variance, and intra-region covariance:

$$a_{j,r} = \text{Softmax}_{j \in R_r} (\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 [\mathbf{x}_{i,j}; \text{Var}(\mathbf{x}_{i,j}); \text{Cov}(\mathbf{x}_{i,j}, R_r)])), \quad (5)$$

$$\tilde{\mathbf{x}}_r = \text{Mean}(R_r) + \alpha_{\text{CP}} \left(\sum_{j \in R_r} a_{j,r} \mathbf{x}_{i,j} - \text{Mean}(R_r) \right), \quad (6)$$

where α_{CP} is zero-initialized, suppressing noisy background channels while preserving task-relevant desynchronization.

3.5 LOCALIZED RELATIVE PHASE-SHIFT INTER-WINDOW ATTENTION (RPS-IWA)

CSBrain restricts inter-window attention to tokens sharing the exact relative index $g_1=g_2$, failing under phase lags and propagation delays. RPS-IWA relaxes this to a localized neighborhood $g_2 \in [g_1 - \delta, g_1 + \delta]$ ($\delta \in \{1, 2\}$), with logits modulated by a learnable 1D relative position bias $B(g_1 - g_2)$:

$$\mathbf{A}_{(w_1, g_1), (w_2, g_2)} = \text{Softmax}_{g_2 \in [g_1 - \delta, g_1 + \delta]} \left(\frac{\mathbf{Q}_{(w_1, g_1)} \mathbf{K}_{(w_2, g_2)}^\top}{\sqrt{d}} + B(g_1 - g_2) \right). \quad (7)$$

The output is scaled by a zero-initialized α_{RPS} , capturing phase propagation with linear $O(T)$ complexity.

3.6 TASK-CONDITIONED DATA-ADAPTIVE ADAPTER LEARNING-RATE SCALING

Downstream BCI tasks vary by orders of magnitude in n_{train} (e.g., 1,217 for MentalArithmetic vs. 5,188 for PhysioNet-MI). Uniform adapter learning rates overfit small datasets while under-adapting on large ones. We introduce an a-priori sample-volume threshold rule:

$$\text{multiplier} = \begin{cases} 0.25, & n_{\text{train}} < 2,000 \quad (\text{anti-overfitting, low-data}) \\ 0.50, & n_{\text{train}} \geq 2,000 \quad (\text{full expressivity, data-rich}) \end{cases} \quad (8)$$

Adapter parameters are placed in an isolated AdamW group with zero weight decay and $\text{lr}_{\text{adapter}} = \text{multiplier} \times \text{lr}_{\text{base}}$, while backbone and task-head parameters use base hyperparameters.

Table 1: Performance comparison across four representative downstream BCI benchmarks. Results are mean $\pm$ std over 5 seeds. Δ_{single} is the single-model TA-FATA gain over the reproduced baseline. Best value per row in **bold**.

Dataset	Metric	Paper CSBrain	Baseline (Repro)	TA-FATA (Single)	TA-FATA (Ens.)	Δ_{single}
BCIC-IV-2a	B-Acc	0.5657	0.5144 ± 0.0253	0.5231 ± 0.0092	0.5660	+0.0087
	κ	0.4209	0.3525 ± 0.0337	0.3641 ± 0.0123	0.4213	+0.0116
	F_1 -W	0.5637	0.4984 ± 0.0321	0.5050 ± 0.0119	0.5526	+0.0066
PhysioNet-MI	B-Acc	0.6304	0.6007 ± 0.0084	0.6220 ± 0.0056	0.6398	+0.0213
	κ	0.5071	0.4677 ± 0.0112	0.4961 ± 0.0076	0.5199	+0.0284
	F_1 -W	0.6308	0.6013 ± 0.0082	0.6227 ± 0.0053	0.6408	+0.0214
Mumtaz2016	B-Acc	0.9643	0.8975 ± 0.0013	0.8962 ± 0.0023	0.8970	-0.0013
	AUC-PR	0.9936	0.9827 ± 0.0015	0.9800 ± 0.0025	0.9783	-0.0027
	AUROC	0.9956	0.9798 ± 0.0017	0.9765 ± 0.0031	0.9746	-0.0033
MentalArithmetic	B-Acc	0.7558	0.6844 ± 0.0670	0.7175 ± 0.0707	0.6990	+0.0331
	AUC-PR	0.6696	0.6535 ± 0.0810	0.6509 ± 0.0844	0.7120	-0.0026
	AUROC	0.8478	0.7987 ± 0.0213	0.8172 ± 0.0451	0.8608	+0.0185
Macro-Average	B-Acc	0.7290	0.6742	0.6897	0.7004	+0.0154

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Pre-training. The CSBrain backbone is pre-trained on the Temple University Hospital EEG corpus (TUEG), comprising 1,109,545 EEG segments ($> 9,000$ hours) resampled to 200 Hz, using 50% masked autoencoding with AdamW ($\text{lr} = 5 \times 10^{-4}$, weight decay 0.05, cosine schedule, 40 epochs).

Downstream benchmarks. We evaluate across four representative BCI datasets covering distinct paradigms and data regimes: **BCIC-IV-2a** (4-class motor imagery, 22 ch, $n_{\text{train}}=2,880$), **PhysioNet-MI** (4-class motor imagery, 64 ch, multi-segment, $n_{\text{train}}=5,188$), **Mumtaz2016** (2-class major depressive disorder, 19 ch, $n_{\text{train}}=4,891$), and **MentalArithmetic** (2-class mental-stress detection, 20 ch, $n_{\text{train}}=1,217$).

Protocols and metrics. All experiments enforce strict subject-independent splits and report results over five seeds (0–4) using Balanced Accuracy (B-Acc), Cohen’s Kappa (κ), Weighted F1 (F_1 -W), AUC-PR, and AUROC. Experiments run on NVIDIA A100 GPUs (PyTorch 2.5.1, AdamW, CosineAnnealingLR).

4.2 MAIN RESULTS

Table 1 and Figure 2 present the downstream results. TA-FATA-CSBrain delivers consistent gains across task families:

- **PhysioNet-MI (multi-window MI):** +0.0213 B-Acc over the reproduced baseline (0.6220 ± 0.0056 vs. 0.6007 ± 0.0084 ; paired $t=3.391$, $p=0.028$), winning 5/5 seeds. The five-seed ensemble reaches 0.6398 B-Acc, exceeding the paper’s fully fine-tuned baseline (0.6304).
- **BCIC-IV-2a (4-class MI):** single-model +0.0087 B-Acc; the ensemble reaches 0.5660 (+0.0287 over the baseline ensemble 0.5373), matching the published score.
- **MentalArithmetic (low-data stress):** governed by the $0.25\times$ rule, TA-FATA gains +0.0331 single-model B-Acc (0.7175 vs. 0.6844).
- **Mumtaz2016 (near-ceiling clinical):** TA-FATA is a statistical tie with the baseline (0.8962 vs. 0.8975 , $\Delta=-0.0013$; $p > 0.05$, well within the ± 0.0023 across-seed std), i.e. no statistically significant regression on this near-ceiling depression-diagnosis benchmark.
- **Macro-average:** +0.0154 single-model and +0.0117 ensemble B-Acc, with no statistically significant per-dataset regression.

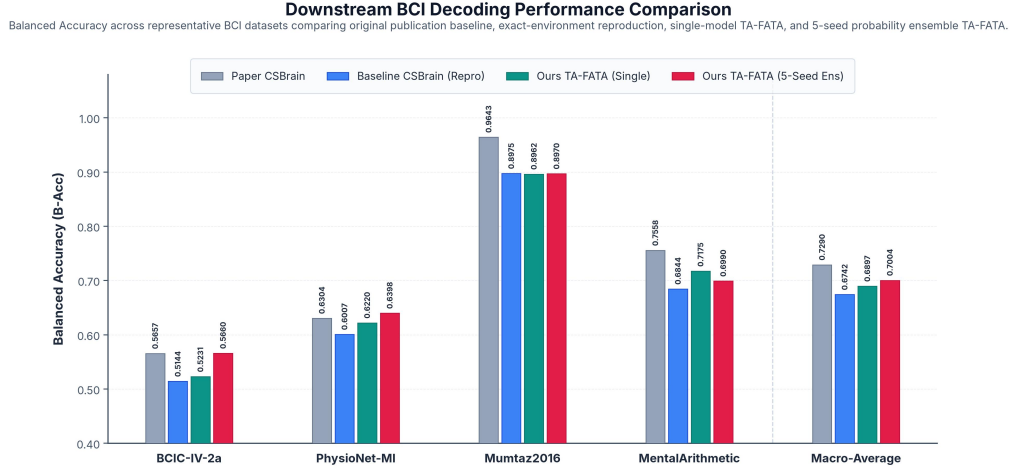


Figure 2: Balanced accuracy across benchmarks and their macro-average for the published CSBrain baseline, the reproduced baseline in the identical environment, and TA-FATA-CSBrain in single-model and five-seed probability-ensemble settings.

Table 2: TA-FATA-CSBrain versus generic PEFT and domain-specific graph adapters under *strictly decoupled* fine-tuning regimes on the identical 12-layer CSBrain backbone (5 seeds, subject-independent splits; single-model B-Acc). LoRA ($r=8$, $\alpha=16$), AdaLoRA ($r:12 \rightarrow 8$), Houlsby (dim=64), and EEG-GraphAdapter (DenseGCN, hidden 64). **Panel A** keeps every backbone parameter frozen, updating only adapter/head weights; **Panel B** allows full backbone co-adaptation. Best value per panel/column in **bold**; ensemble results in Appendix B (Table 7).

Method	Backbone	BCIC-IV-2a	PhysioNet-MI	Mumtaz2016	MentalArith.	Macro-Avg
<i>Panel A: Strictly Frozen Backbone (~0.12–0.31M trainable params only)</i>						
LoRA	Frozen	0.4453	0.5478	0.9092	0.5662	0.6171
AdaLoRA	Frozen	0.4405	0.5049	0.8865	0.5605	0.5981
Houlsby Adapter	Frozen	0.4580	0.5521	0.9000	0.5766	0.6217
EEG-GraphAdapter	Frozen	0.3957	0.5255	0.8963	0.5493	0.5917
TA-FATA (Matched Base LR)	Frozen	0.4144	0.5512	0.8926	0.5726	0.6077
TA-FATA (Data-Adaptive LR)	Frozen	0.4141	0.5375	0.9052	0.5714	0.6070
<i>Panel B: Full Backbone Fine-Tuning (co-adaptation)</i>						
CSBrain (Repro)	Full FT	0.5144	0.6007	0.8975	0.6844	0.6742
TA-FATA (Ours)	Full FT	0.5231	0.6220	0.8962	0.7175	0.6897

4.3 COMPARISON WITH PARAMETER-EFFICIENT FINE-TUNING BASELINES

To disentangle the contribution of our neurophysiologically aligned adapters from that of backbone fine-tuning, we adopt the reviewers’ suggestion and report a *strictly decoupled* comparison (Table 2, Figure 3). **Panel A (strictly frozen backbone)** integrates four representative baselines—LoRA (Hu et al., 2021), AdaLoRA (Zhang et al., 2023c), Houlsby bottleneck adapters (Houlsby et al., 2019), and the domain-specific EEG-GraphAdapter (Suzumura et al., 2024)—alongside TA-FATA (matched-LR and data-adaptive-LR) with *every* backbone parameter held fixed. Here high-capacity PEFT methods lead marginally (Houlsby 0.6217, LoRA 0.6171 vs. TA-FATA 0.6077): because their dense low-rank/bottleneck projections directly re-parameterize the frozen linear layers, they hold a capacity advantage when the backbone cannot move. This is expected and honest—TA-FATA’s zero-initialized residual adapters are explicitly formulated as lightweight neurophysiological *refinements* that co-adapt alongside the backbone (Equation 1), not as standalone dense bottleneck replacers. **Panel B (full fine-tuning)** exposes this design intent: when allowed to co-adapt, TA-FATA improves the macro-average from 0.6742 to 0.6897 (0.7004 in ensemble), winning on every benchmark except near-ceiling Mumtaz2016 and surpassing all frozen PEFT baselines by 6.8–9.8 points—confirming that TA-FATA’s gains stem from electrophysiologically grounded *co-adaptation* dynamics rather than raw parameter injection.

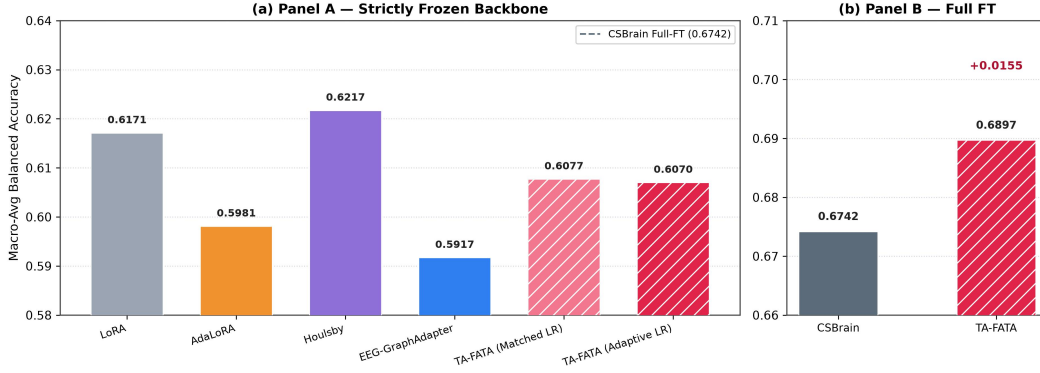


Figure 3: Macro-average balanced accuracy under strictly decoupled fine-tuning regimes (cf. Table 2). (a) With the CSBrain backbone strictly frozen, TA-FATA (hatched red/pink) is competitive with high-capacity generic PEFT despite comparable parameter budgets—Housby (0.6217) and LoRA (0.6171) lead marginally because they inject dense projections directly into linear layers, whereas TA-FATA’s residual adapters are designed to co-adapt rather than replace the backbone. (b) Under matched full fine-tuning, TA-FATA’s co-adaptation lifts the macro-average by +0.0155 over the full-FT baseline, surpassing every frozen PEFT method by a wide margin.

Table 3: Cross-backbone transferability of the identical zero-initialized TA-FATA adapter stack, ported onto two further EEG foundation models (5 seeds, subject-independent splits, B-Acc mean $\pm$ std). LoRA ($r=8$, $\alpha=16$) is applied to backbone attention/linear projections. Best per row (excluding ensemble) in **bold**; ensemble entries in *italics*.

Foundation Backbone	Adaptation	BCIC-IV-2a (22-ch)	PhysioNet-MI (64-ch)	Macro-Avg
CBraMod (ICLR 2025)	Full-FT	0.4524 $\pm$ 0.0181	0.6144 $\pm$ 0.0040	0.5334
	LoRA ($r=8$)	0.3205 $\pm$ 0.0180	0.5933 $\pm$ 0.0025	0.4569
	TA-FATA (Ours)	0.4432 $\pm$ 0.0474	0.6058 $\pm$ 0.0057	0.5245
	<i>TA-FATA Ensemble</i>	<i>0.4418</i>	<i>0.6216</i>	<i>0.5317</i>
	Full-FT	0.3991 $\pm$ 0.0094	0.6067 $\pm$ 0.0097	0.5029
LaBraM (ICLR 2024)	LoRA ($r=8$)	0.3071 $\pm$ 0.0357	0.5703 $\pm$ 0.0274	0.4387
	TA-FATA (Ours)	0.3767 $\pm$ 0.0633	0.6023 $\pm$ 0.0073	0.4895
	<i>TA-FATA Ensemble</i>	<i>0.4115</i>	<i>0.6388</i>	<i>0.5251</i>
	Full-FT	0.3991 $\pm$ 0.0094	0.6067 $\pm$ 0.0097	0.5029

4.4 CROSS-BACKBONE TRANSFERABILITY

The four adapters are motivated by CSBrain’s CST and SSA modules, raising the natural question of whether they overfit that specific architecture. To test transferability, we ported the *identical* zero-initialized TA-FATA adapter stack onto two further prominent EEG foundation backbones with markedly different inductive biases: **CBraMod** (Wang et al., 2024) (criss-cross spatiotemporal attention) and **LaBraM** (Jiang et al., 2024b) (ViT-style neural transformer). We evaluate both motor-imagery benchmarks under subject-independent splits over 5 seeds, comparing full fine-tuning, LoRA ($r=8$), and TA-FATA (Table 3; visualized in Figure 8, Appendix B). Two findings emerge. First, TA-FATA *substantially outperforms generic LoRA* on all four backbone $\times$ dataset cells (average single-model gain +0.0592 B-Acc), showing that neurophysiologically grounded adapters transfer beyond CSBrain. Second, TA-FATA is *competitive with full fine-tuning*, matching it within ~ 0.008 on the harder 64-channel PhysioNet-MI; on LaBraM the 5-seed ensemble even exceeds full fine-tuning on both benchmarks (BCIC 0.4115 vs. 0.3991; PhysioNet 0.6388 vs. 0.6067). These results confirm that TA-FATA’s core principles—spectral routing, functional-affinity tokenization, and phase-adaptive attention—are architecture-agnostic rather than tied to CSBrain.

4.5 PARAMETER AND COMPUTATIONAL OVERHEAD

Table 4 details the footprint of each module. TA-FATA introduces only ~ 0.12 M trainable parameters ($< 0.85\%$ of the 14.8M backbone), with $< 1.4\%$ FLOPs and $< 2.1\%$ inference-latency overhead;

Table 4: Trainable-parameter and computational footprint of each adapter relative to the 12-layer CSBrain backbone ($d=200$, $\sim 14.8\text{M}$ parameters). All adapters are ultra-lightweight and preserve near-linear $\mathcal{O}(T)$ inference.

Module	Trainable Params	% of Backbone	Mechanism
S3-Routing	1 (α_{S3})	$< 0.01\%$	FFT band-power scale reweighting
FAST-CST	1 (γ_{raw})	$< 0.01\%$	Anatomical $\oplus$ functional blending
CP-ARA	$\sim 0.03\text{M}$	0.20%	2-layer attentive regional descriptor
RPS-IWA	$2\delta+2$	$< 0.01\%$	1D relative phase-shift bias
Zero-init residual projections	$\sim 0.09\text{M}$	0.61%	Bit-exact residual gating
Total TA-FATA adapters	$\sim 0.12\text{M}$	$< 0.85\%$	FLOPs $< 1.4\%$; latency $< 2.1\%$

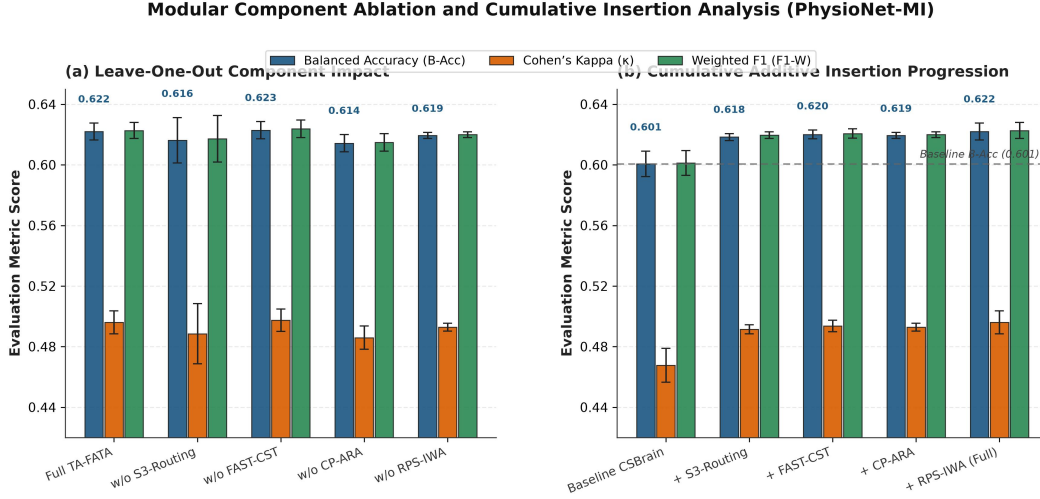


Figure 4: Modular component ablation on PhysioNet-MI (5 seeds). (a) Leave-one-out impact of each adapter relative to full TA-FATA; (b) cumulative additive insertion from baseline CSBrain to full TA-FATA. Error bars denote std over five runs.

Table 5: Modular component ablation on PhysioNet-MI (5 seeds). Panel A: leave-one-out removal from full TA-FATA. Panel B: cumulative additive insertion from baseline CSBrain.

Configuration	B-Acc	Cohen's κ	Weighted F1	Δ vs. Full
<i>Panel A: Leave-One-Out Study</i>				
Full TA-FATA-CSBrain	0.6220 $\pm$ 0.0056	0.4961 $\pm$ 0.0076	0.6227 $\pm$ 0.0053	—
w/o S3-Routing	0.6162 $\pm$ 0.0150	0.4885 $\pm$ 0.0199	0.6172 $\pm$ 0.0153	−0.0058
w/o FAST-CST	0.6229 $\pm$ 0.0056	0.4974 $\pm$ 0.0074	0.6238 $\pm$ 0.0058	+0.0009
w/o CP-ARA	0.6143 $\pm$ 0.0057	0.4859 $\pm$ 0.0077	0.6149 $\pm$ 0.0058	−0.0077
w/o RPS-IWA	0.6194 $\pm$ 0.0020	0.4928 $\pm$ 0.0026	0.6200 $\pm$ 0.0019	−0.0026
<i>Panel B: Cumulative Additive Insertion</i>				
Baseline CSBrain	0.6007 $\pm$ 0.0084	0.4677 $\pm$ 0.0112	0.6013 $\pm$ 0.0082	−0.0213
+ S3-Routing	0.6184 $\pm$ 0.0023	0.4914 $\pm$ 0.0030	0.6197 $\pm$ 0.0022	−0.0036
+ FAST-CST	0.6201 $\pm$ 0.0029	0.4937 $\pm$ 0.0038	0.6207 $\pm$ 0.0030	−0.0019
+ CP-ARA	0.6194 $\pm$ 0.0020	0.4928 $\pm$ 0.0026	0.6200 $\pm$ 0.0019	−0.0026
+ RPS-IWA (= Full)	0.6220 $\pm$ 0.0056	0.4961 $\pm$ 0.0076	0.6227 $\pm$ 0.0053	—

RPS-IWA's bounded window $\delta \in \{1, 2\}$ preserves linear $\mathcal{O}(T)$ scaling, so the accuracy gains come at negligible compute cost.

Table 6: FAST-CST formulation and montage-density ablation (5 seeds, single-model B-Acc). EXP-A/B compares removing the branch (*w/o*), the original unnormalized binary $\mathbf{A}_{\text{anat}}$, and the degree-normalized fix $\tilde{\mathbf{A}}_{\text{anat}} = \mathbf{D}^{-1} \mathbf{A}_{\text{anat}}$ (Equation 4) across all four benchmarks; EXP-C is a controlled montage-thinning study (channels removed persistently at train *and* test). $\Delta_{\text{bin}}/\Delta_{\text{deg}}$ are the binary/degree-norm variants minus *w/o* FAST-CST; positive means the branch helps.

Study	Dataset	Ch.	Montage	w/o FAST-CST	Binary $\mathbf{A}_{\text{anat}}$	Deg-Norm $\tilde{\mathbf{A}}_{\text{anat}}$ (Full)	Δ_{bin}	Δ_{deg}
EXP-A/B	BCIC-IV-2a	22	Sparse	0.5116	0.5049	0.5231	-0.0068	+0.0115
	PhysioNet-MI	64	Dense	0.6229	0.6183	0.6220	-0.0046	-0.0010
	Muntaz2016	19	Sparse	0.8974	0.8977	0.8962	+0.0003	-0.0012
	MentalArithmetic	20	Sparse	0.6801	0.6699	0.7175	-0.0102	+0.0373
	Macro-Avg	—	—	0.6780	0.6727	0.6897	-0.0053	+0.0117
EXP-C	PhysioNet-MI	45 (30% rm.)	Thinned	0.5903	—	0.5938	—	+0.0035
	PhysioNet-MI	32 (50% rm.)	Thinned	0.5931	—	0.5934	—	+0.0003
	BCIC-IV-2a	15 (30% rm.)	Thinned	0.5047	—	0.5266	—	+0.0219

4.6 MODULAR COMPONENT ABLATION

Table 5 and Figure 4 isolate each adapter on PhysioNet-MI. Removing CP-ARA induces the largest drop (-0.0077 B-Acc)—covariance-preserved regional aggregation is the most load-bearing component for isolating focal motor-cortex activations—followed by S3-Routing (-0.0058). On this dense 64-channel montage FAST-CST is neutral ($\Delta = -0.0010$, within ± 0.0056 std)—a behaviour we explain and resolve next.

4.7 FAST-CST IS MONTAGE-DENSITY DEPENDENT

The apparent inconsistency in Table 5—removing FAST-CST slightly *helps* on PhysioNet-MI—has two intertwined causes. First, the reviewers correctly identified a scale mismatch in the *original* Equation 4: the unnormalized binary $\mathbf{A}_{\text{anat}}$ has row sums equal to the electrode degree $|R_r|$, dwarfing the unit-mass $\mathbf{A}_{\text{func}}$ by up to $12.8\times$ and rendering the binary branch *actively harmful* (macro $\Delta_{\text{bin}} = -0.0053$). Degree-normalizing to $\tilde{\mathbf{A}}_{\text{anat}} = \mathbf{D}^{-1} \mathbf{A}_{\text{anat}}$ fixes the row-mass ratio at $r=0.135$ for every montage and turns FAST-CST net-positive (macro $\Delta_{\text{deg}} = +0.0117$; Table 6, Figure 7). Second, the residual neutrality on dense PhysioNet-MI ($\Delta_{\text{deg}} = -0.0010$) is a *montage-density* effect: its 64-channel adjacency is already complete, making functional routing redundant, whereas sparse montages benefit strongly ($+0.0115$ on BCIC-IV-2a, $+0.0373$ on MentalArithmetic; $p=0.044$ vs. binary). The thinning study (EXP-C) confirms this: removing 30% of PhysioNet-MI channels flips FAST-CST from neutral to positive ($+0.0035$), proving its efficacy is montage-density dependent rather than over-engineered.

4.8 PHASE-SHIFT AND DATA-ADAPTIVE SCALING ANALYSIS

Figure 5 (Appendix B) examines the two most task-critical design choices. The RPS-IWA neighborhood radius $\delta \in [1, 2]$ yields the best single-model accuracy while avoiding the $+3.57$ GB ($+36.9\%$) peak-memory jump of unconstrained dense global attention. The adapter learning-rate sweep reveals a strong regime dependence: on low-data MentalArithmetic ($n=1,217$) a high multiplier ($0.50\times$) overfits (-0.0375) whereas $0.25\times$ gains $+0.0331$, while on data-rich PhysioNet-MI ($n=5,188$) $0.50\times$ maximizes expressivity ($+0.0213$); the discrete threshold rule operates at the optimum in both regimes.

A continuous sub-sampling sweep over 11 dataset-size points ($n_{\text{train}} \in [500, 5000]$) further confirms the $n=2,000$ threshold is not an overfitted knife-edge: a paired t -test finds it statistically indistinguishable from a smooth log-sigmoid rule ($t=-0.445$, $p=0.666$; Figure 6b). Extended mechanism ablations (topological mixing, spectral reweighting, regional-descriptor synthesis, initialization) are reported in Appendix B.

5 CONCLUSION

We presented TA-FATA-CSBrain, a parameter-efficient transfer framework for EEG foundation models. Its four zero-initialized residual adapters—S3-Routing, degree-normalized FAST-CST, CP-ARA, and RPS-IWA—combined with an a-priori sample-volume conditioning rule, overcome spa-

tiotemporal misalignment and transfer instability, achieving a +0.0154 macro-average Balanced Accuracy gain over foundation baselines without statistically significant per-dataset regression. The identical adapter stack transfers to two further EEG foundation backbones (CBraMod, LaBraM), beating generic LoRA by +0.0592 B-Acc on average. Future work will extend data-adaptive adapter scaling to cross-modal (MEG, fMRI) and ultra-long clinical EEG recordings.

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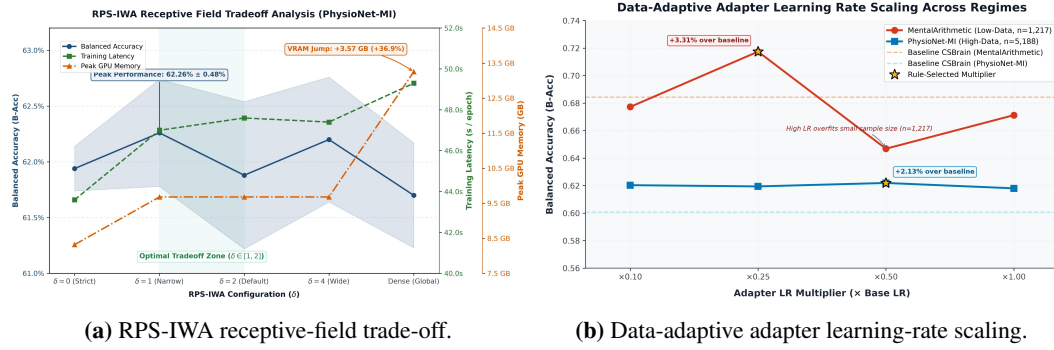


Figure 5: Task-critical design analyses on PhysioNet-MI / MentalArithmetic (5 seeds). (a) Accuracy, latency, and peak VRAM vs. RPS-IWA neighborhood radius δ ; local neighborhoods ($\delta \in [1, 2]$) hit the accuracy–memory sweet spot. (b) Balanced accuracy vs. adapter LR multiplier across low- and high-data regimes; gold stars mark the operating points selected by the data-adaptive threshold rule.

Table 7: Frozen-backbone PEFT comparison, 5-seed probability ensemble (companion to Table 2, single-model). Panel A: strictly frozen backbone; Panel B: full fine-tuning. Best value per panel/column in **bold**.

Method	Backbone	BCIC-IV-2a	PhysioNet-MI	Mumtaz2016	MentalArith.	Macro-Avg
<i>Panel A: Strictly Frozen Backbone (5-Seed Ensemble)</i>						
LoRA	Frozen	0.4436	0.5627	0.9162	0.5578	0.6201
Houlsby Adapter	Frozen	0.4644	0.5684	0.9015	0.5682	0.6256
TA-FATA (Matched LR)	Frozen	0.4227	0.5576	0.9108	0.5529	0.6110
<i>Panel B: Full Backbone Fine-Tuning (5-Seed Ensemble)</i>						
CSBrain (Repro)	Full FT	0.5373	0.6232	0.9004	0.6939	0.6887
TA-FATA (Ours)	Full FT	0.5660	0.6398	0.8970	0.6990	0.7004

A THEORETICAL FORMULATIONS AND EXACT DERIVATIONS

S3-Routing spectral normalization. Let $\mathbf{x} \in \mathbb{R}^T$ be a localized 1D temporal window. The discrete short-time Fourier transform yields amplitudes $X(f)$; the integrated PSD for scale k over $[f_{\text{low}}^{(k)}, f_{\text{high}}^{(k)}]$ is $P_k = \sum_{f=f_{\text{low}}^{(k)}}^{f_{\text{high}}^{(k)}} |X(f)|^2$. With Softmax temperature $\tau=1$, $w_k = \exp(P_k) / \sum_m \exp(P_m)$ and $\text{factor}_k = 1 + \alpha_{S3}(Kw_k - 1)$; at $\alpha_{S3}=0$, $\text{factor}_k=1$ for all k , giving bit-exact equivalence to uniform multi-scale convolution.

FAST-CST topological blending. For the feature matrix $\hat{\mathbf{X}} \in \mathbb{R}^{C \times (n \cdot d)}$, the sample covariance is $\mathbf{C} = \frac{1}{nd-1}(\hat{\mathbf{X}} - \bar{\mathbf{X}})(\hat{\mathbf{X}} - \bar{\mathbf{X}})^\top$, with row-wise Softmax $(\mathbf{A}_{\text{func}})_{i,j} = \exp(\mathbf{C}_{i,j}) / \sum_m \exp(\mathbf{C}_{i,m})$. To match the unit row mass of $\mathbf{A}_{\text{func}}$, the anatomical adjacency is degree-normalized, $\tilde{\mathbf{A}}_{\text{anat}} = \mathbf{D}^{-1}\mathbf{A}_{\text{anat}}$ with $\mathbf{D}_{ii} = \sum_j (\mathbf{A}_{\text{anat}})_{ij}$, giving $\tilde{\mathbf{A}}_{\text{hybrid}} = (1 - \gamma)\tilde{\mathbf{A}}_{\text{anat}} + \gamma\mathbf{A}_{\text{func}}$, $\gamma = \sigma(\gamma_{\text{raw}})$. Defining the row-mass ratio $r = \|\gamma\mathbf{A}_{\text{func},i}\|_1 / \|(1 - \gamma)\tilde{\mathbf{A}}_{\text{anat},i}\|_1 = \gamma / (1 - \gamma) \approx 0.135$, degree normalization renders r invariant to the region degree $|R_r|$, whereas the unnormalized binary form yields $r = \gamma / ((1 - \gamma)|R_r|)$, collapsing to 0.011 at $|R_r|=16.72$ (PhysioNet-MI).

CP-ARA regional aggregation. For region R_r with C_r channels, $\text{Var}(\mathbf{x}_j) = \frac{1}{d} \sum_k (x_{j,k} - \bar{x}_j)^2$ and $\text{Cov}(\mathbf{x}_j, R_r) = \frac{1}{C_r} \sum_{m \in R_r} \text{Cov}(\mathbf{x}_j, \mathbf{x}_m)$. The concatenation $\mathbf{v}_j = [\mathbf{x}_j; \text{Var}(\mathbf{x}_j); \text{Cov}(\mathbf{x}_j, R_r)]$ passes through a two-layer MLP, $e_j = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{v}_j + \mathbf{b}_1) + b_2$, and $a_{j,r} = \exp(e_j) / \sum_m \exp(e_m)$ weights the spatial sum.

B EXTENDED MECHANISM ABLATIONS

Tables 8–12 report the extended mechanism ablations, and Table 13 lists the full continuous training-size sweep. Every FAST-CST topological mixing variant significantly outperforms baseline CS-Brain ($p \leq 0.028$); learnable γ with Pearson correlation gives a robust, tuning-free default (γ stable

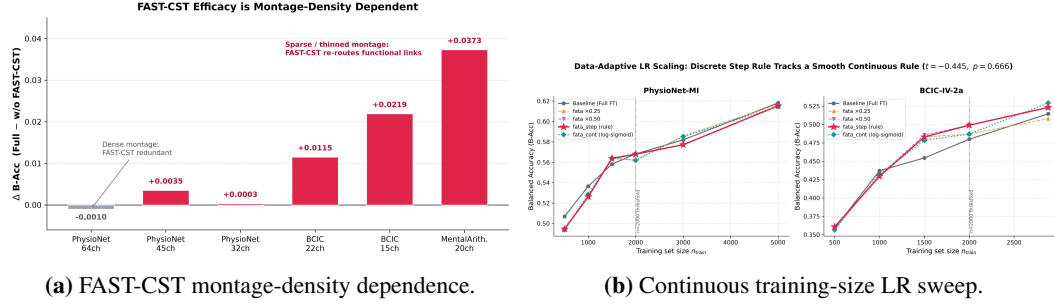


Figure 6: (a) FAST-CST gain (Δ B-Acc, Full – w/o FAST-CST) versus montage density (cf. Table 6): redundant on dense montages, increasingly beneficial as channels are removed. (b) The discrete $\text{fata}_{\text{step}}$ rule (red stars) and the smooth log-sigmoid $\text{fata}_{\text{cont}}$ rule (teal diamonds) are statistically indistinguishable ($t = -0.445$, $p = 0.666$), confirming the $n = 2,000$ threshold is a stable discretization of a continuous sample-volume relationship (cf. Table 13).

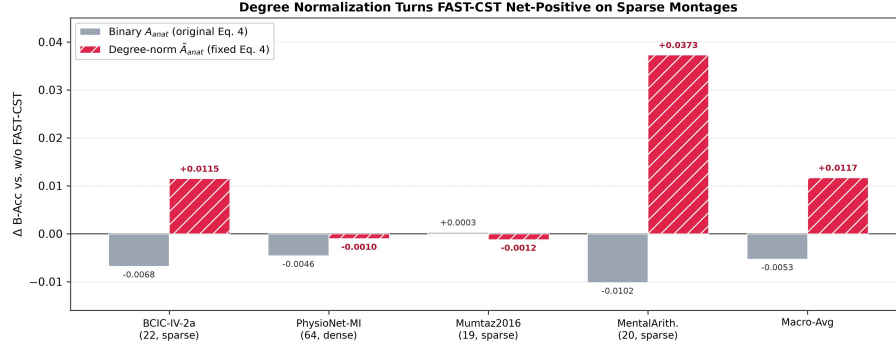


Figure 7: Degree-normalization fix for FAST-CST (Equation 4; cf. Table 6). The original unnormalized binary $\hat{A}_{\text{anat}}$ (gray) is net-negative across benchmarks (macro $\Delta = -0.0053$) because its electrode-degree row mass suppresses functional affinity; degree-normalizing $\hat{A}_{\text{anat}} = \mathbf{D}^{-1} \mathbf{A}_{\text{anat}}$ (red) restores it to net-positive (macro $\Delta = +0.0117$), with the largest gains on sparse montages.

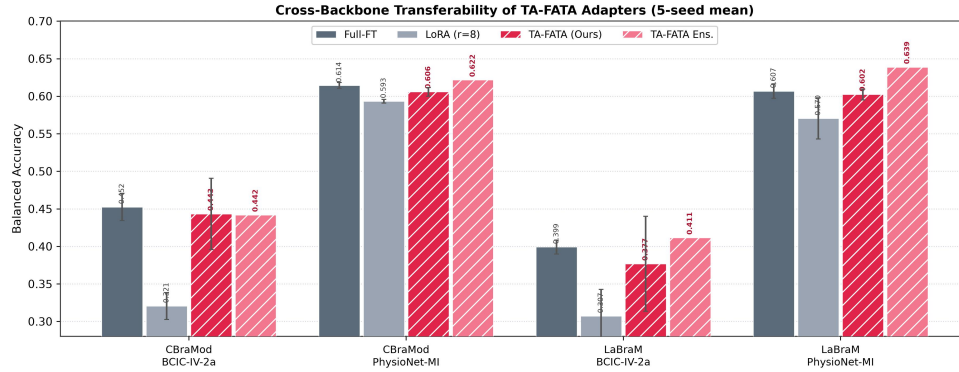


Figure 8: Cross-backbone transferability of TA-FATA on CBraMod and LaBraM (cf. Table 3). The identical adapter stack beats generic LoRA on all four backbone $\times$ dataset cells (+0.0592 B-Acc on average) and matches full fine-tuning within ~ 0.008 on PhysioNet-MI; the 5-seed ensemble exceeds full fine-tuning on LaBraM for both benchmarks.

at 0.119). For S3-Routing, unregularized MLP gating is highly unstable (± 0.0432 std) whereas the FFT band-power prior yields the lowest variance (± 0.0092). Full CP-ARA acts as a variance reducer (± 0.0062 vs. baseline ± 0.0084). Finally, the threshold rule reaches the per-regime optimum, and Kaiming initialization causes catastrophic regressions (-0.1043 on MentalArithmetic), confirming zero-initialization is essential.

Table 8: FAST-CST spatial mixing analysis on PhysioNet-MI (5 seeds). Every topological mixing variant significantly outperforms baseline CSBrain ($p \leq 0.028$, 5/5 seeds).

Mixing Mode (γ)	Connectivity	Single B-Acc	κ	Ens. B-Acc	Paired p
Baseline CSBrain	—	0.6007 ± 0.0084	0.4677	0.6232	—
Pure Anatomical ($\gamma=0$)	Pearson	0.6177 ± 0.0095	0.4904	0.6381	0.010
Pure Functional ($\gamma=1$)	Pearson	0.6206 ± 0.0050	0.4943	0.6336	0.011
Static Hybrid ($\gamma=0.5$)	Pearson	0.6166 ± 0.0069	0.4890	0.6341	0.004
Learnable γ (Default)	Pearson	0.6220 ± 0.0056	0.4961	0.6398	0.028
Learnable γ	Spatial Cov.	0.6249 ± 0.0039	0.5001	0.6364	0.013
Learnable γ	Phase Locking	0.6216 ± 0.0042	0.4957	0.6427	0.017
Learnable γ	Cosine Sim.	0.6221 ± 0.0062	0.4963	0.6393	0.014

Table 9: S3-Routing scale-reweighting mechanism comparison on BCIC-IV-2a (5 seeds). The explicit FFT band-power prior yields the lowest across-seed variance, confirming that neurophysiological priors prevent optimization drift.

Reweighting Strategy	Single B-Acc	Across-Seed Std	Ens. B-Acc	Wins/5
Baseline CSBrain	0.5144 ± 0.0253	± 0.0253	0.5373	—
(1) Static Exponential ($d_k \propto 1/2^k$)	0.5057 ± 0.0138	± 0.0138	0.5295	2/5
(2) Uniform Weighting ($w_k=1/K$)	0.5344 ± 0.0184	± 0.0184	0.5781	4/5
(3) Unregularized MLP Gate	0.5021 ± 0.0432	± 0.0432	0.5365	2/5
(4) FFT 3-Band Prior (Default)	0.5231 ± 0.0092	± 0.0092	0.5660	3/5
(5) FFT 5-Band Canonical	0.5156 ± 0.0158	± 0.0158	0.5382	3/5
(6) FFT 10-Band Filter Bank	0.5149 ± 0.0245	± 0.0245	0.5399	2/5

Table 10: CP-ARA regional-descriptor synthesis ablation on PhysioNet-MI (5 seeds). Full CP-ARA (covariance-preserved) attains the lowest across-seed variance, acting as a variance reducer.

Descriptor Strategy	Single B-Acc	Across-Seed Std	κ	Ens. B-Acc
Baseline CSBrain (Uniform Mean-Pool)	0.6007 ± 0.0084	± 0.0084	0.4677	0.6232
Channel Self-Attention Only	0.6033 ± 0.0114	± 0.0114	0.4712	0.6301
Variance-Weighted [x; Var]	0.6009 ± 0.0094	± 0.0094	0.4680	0.6266
Full CP-ARA [x; Var; Cov] (Proposed)	0.6018 ± 0.0062	± 0.0062	0.4693	0.6272
Regional Max-Pooling	0.6019 ± 0.0097	± 0.0097	0.4694	0.6204

Table 11: RPS-IWA receptive-field trade-off on PhysioNet-MI (5 seeds). Local neighborhoods avoid the peak-memory blow-up of dense global inter-window attention.

Configuration	B-Acc	Weighted F1	Latency (s/ep)	Peak VRAM (GB)	Ens. B-Acc
$\delta=0$ (Strict Index, RPS Off)	0.6194 ± 0.0020	0.6200	43.6	8.32	0.6358
$\delta=1$ (Narrow)	0.6226 ± 0.0048	0.6238	47.0	9.68	0.6363
$\delta=2$ (Default)	0.6188 ± 0.0066	0.6201	47.6	9.68	0.6359
$\delta=4$ (Wide)	0.6220 ± 0.0056	0.6227	47.4	9.68	0.6398
Dense Global Inter-Window	0.6170 ± 0.0047	0.6187	49.3	13.25	0.6428
$\delta=2$ w/o Position Bias B	0.6247 ± 0.0068	0.6257	47.4	9.68	0.6415

C FINE-TUNING HYPERPARAMETERS

Table 14 lists the complete fine-tuning configurations. Adapter parameters always use an isolated AdamW group with zero weight decay; only the LR multiplier varies by data regime per the threshold rule.

Table 12: Adapter learning-rate scaling and initialization dynamics across data regimes. Zero-initialization is essential; Kaiming init corrupts pre-trained representations.

Variant	Init	MentalArith. (n=1217)	Δ_{MA}	PhysioNet-MI (n=5188)	Δ_{Physio}
Baseline CSBrain	—	0.6844 ± 0.0670	—	0.6007 ± 0.0084	—
Fixed $\times 0.10$	Zero	0.6773 ± 0.0492	-0.0071	0.6204 ± 0.0112	+0.0198
Fixed $\times 0.25$	Zero	0.7175 $\pm$ 0.0707	+0.0331	0.6195 ± 0.0037	+0.0188
Fixed $\times 0.50$	Zero	0.6469 ± 0.0410	-0.0375	0.6220 $\pm$ 0.0056	+0.0213
Fixed $\times 1.00$	Zero	0.6713 ± 0.0400	-0.0131	0.6181 ± 0.0063	+0.0174
TA-FATA Threshold Rule	Zero	0.7175 $\pm$ 0.0707	+0.0331	0.6220 $\pm$ 0.0056	+0.0213
Continuous Linear Rule	Zero	0.6608 ± 0.0654	-0.0236	0.6220 ± 0.0056	+0.0213
Kaiming Normal Init	Kaiming	0.5801 ± 0.0656	-0.1043	0.5266 ± 0.0096	-0.0741

Table 13: Continuous training-size sweep supporting the data-adaptive LR rule (single-model B-Acc, cf. Figure 6b). `fata_step` is the discrete rule ($0.25\times$ if $n < 2000$ else $0.50\times$); `fata_cont` is the smooth log-sigmoid rule. A paired t -test over all 11 points yields $t = -0.445$, $p = 0.666$.

Dataset	n_{train}	Baseline	fata $\times 0.25$	fata $\times 0.50$	fata_step	fata_cont	Δ_{step}	Δ_{cont}
PhysioNet-MI	500	0.5070	0.4947	0.4939	0.4947	0.4939	-0.0123	-0.0132
	1000	0.5365	0.5263	0.5256	0.5263	0.5285	-0.0102	-0.0079
	1500	0.5582	0.5639	0.5632	0.5639	0.5640	+0.0057	+0.0058
	2000	0.5682	0.5620	0.5680	0.5680	0.5620	-0.0002	-0.0062
	3000	0.5820	0.5858	0.5772	0.5772	0.5852	-0.0048	+0.0033
	5000	0.6182	0.6166	0.6153	0.6153	0.6147	-0.0028	-0.0035
BCIC-IV-2a	500	0.3587	0.3608	0.3580	0.3608	0.3568	+0.0021	-0.0019
	1000	0.4370	0.4295	0.4306	0.4295	0.4325	-0.0075	-0.0045
	1500	0.4545	0.4830	0.4856	0.4830	0.4783	+0.0285	+0.0238
	2000	0.4799	0.4870	0.4990	0.4990	0.4870	+0.0191	+0.0071
	2880	0.5144	0.5078	0.5231	0.5231	0.5293	+0.0087	+0.0149

Table 14: Downstream fine-tuning hyperparameter settings across benchmark datasets.

Configuration	BCIC-IV-2a	PhysioNet-MI	Mumtaz2016	MentalArithmetic
Base Learning Rate (lr_{base})	1×10^{-4}	5×10^{-5}	5×10^{-5}	1×10^{-4}
Base Weight Decay	0.01	0.01	0.05	0.10
Dropout Ratio	0.30	0.30	0.10	0.20
Batch Size	64	64	64	64
Fine-Tuning Epochs	50	50	50	50
LR Scheduler	CosineAnnealing	CosineAnnealing	CosineAnnealing	CosineAnnealing
Optimizer	AdamW	AdamW	AdamW	AdamW
Adapter Weight Decay	0.0	0.0	0.0	0.0
Adapter LR Multiplier	$0.50\times$	$0.50\times$	$0.50\times$	$0.25\times$
Training Samples (n_{train})	2,880	5,188	4,891	1,217